1. Clinical Question & Core Hypothesis

Clinical Question

SARIFA-positive and TGF-β–high MIBC tumors exhibit pronounced stromal remodeling yet show heterogeneous benefit from immunotherapy — why does a stromally inflamed/ remodeled microenvironment sometimes prevent, rather than permit, effective anti-tumor immune responses? 2) Tier 1 paths used: (a) TGF-β as a fibrogenic, immunosuppressive driver (core mechanistic mediator of ECM deposition and immune exclusion) and (b) SARIFA (tumor–adipocyte/ stromal interaction phenotype) as the microenvironmental readout that captures fibro-adipocyte remodeling linked to invasion and altered immune contexture. I built on these because together they provide a mechanistic axis (cytokine-driven stromal remodeling → SARIFA microenvironment) that plausibly blocks immune effector access. 3) Tier 2 support: pZVR, PSA and metastasis-linked signals reinforce the clinical consequence (nodal spread / worse outcome) of the stromal state; SDC2/TFPI2 methylation and AI/MSI-pathways illustrate noninvasive and computational routes to measure/scale biomarkers and validate the stromal signal; these support testable stratification and deployment strategies. 4) Why therapy_response as endpoint: the core translational problem is immunotherapy failure in a subset of MIBC despite apparent microenvironmental changes — demonstrating a mechanistic cause of checkpoint blockade resistance has immediate therapeutic implications (patient selection, combination trials with anti–TGF-β or stroma-targeting agents). 5) Deliberate exclusions: I excluded adding TLS/TIL density as a core imaging variable and PSA/SDC2/TFPI2/pZVR/MSI/AI as anchor genes to keep the hypothesis sharply focused on the stromal mechanism (TSR + TGFB1 + SARIFA). Those excluded items are listed as secondary_modifiers because they are valuable for validation, measurement, or staging but would dilute the mechanistic claim if placed in the core hypothesis?

Working Hypothesis

High stromal-to-tumor ratio (TSR) in MIBC, when accompanied by elevated TGFB1 signaling and a SARIFA-positive fibro-adipocyte remodeling phenotype, creates both a physical stromal barrier and a suppressive biochemical milieu that limit effector T cell infiltration, TLS formation, and anti-tumor function, thereby reducing response to PD-1/PD-L1 blockade. This effect is most pronounced in tumors with TSR-high/TGFB1-high/SARIFA-positive status and is predictive of poor objective response rate and shorter progression-free and overall survival on immunotherapy compared with TSR-low or TGFB1-low/SARIFA-negative tumors. Importantly, these tumors may be resensitized by combination therapy that disrupts TGF-β signaling or remodels the stroma (e.g., anti–TGF-β agents, CAF-targeting strategies) in addition to immune checkpoint blockade. A composite biomarker score integrating TSR (imaging/histology), TGFB1 pathway activity, and SARIFA status will stratify patients and guide therapeutic decision-making, enabling testing of combination regimens in prospective trials.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB1, SARIFA
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1 + SARIFA).

SARIFA: SARIFA-positive and TGF-β–high MIBC tumors exhibit pronounced stromal remodeling yet show heterogeneous benefit from immunotherapy — why does a stromally inflamed/ remodeled microenvironment sometimes prevent, rather than permit, effective anti-tumor immune responses? 2) Tier 1 paths used: (a) TGF-β as a fibrogenic, immunosuppressive driver (core mechanistic mediator of ECM deposition and immune exclusion) and (b) SARIFA (tumor–adipocyte/ stromal interaction phenotype) as the microenvironmental readout that captures fibro-adipocyte remodeling linked to invasion and altered immune contexture.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	PSA	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	AI-based biomarker pipelines	Supporting modifier from knowledge graph
Secondary	TAF (tumor-associated fibroblast signatures)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: establish TSR as an imaging/histology-derived proxy for stromal remodeling;

Prediction 2: integrate TGFB1 pathway activity and SARIFA status to form a mechanistic composite biomarker;

Prediction 3: test the hypothesis across retrospective and prospective cohorts with multi-omic and spatial profiling;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- pZVR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PSA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.